

IP Addressing & Subnetting Practice Worksheet

Computer Networks | Classful & Subnetted IP Address Problems (Class A, B & C)

Instructions: For each of the given IP address / prefix combinations below, determine the network address, subnet mask, number of host bits available, total number of addresses, number of usable host addresses, the first and last usable host IP addresses, and the broadcast address. Show your working.

Question 1		192.168.1.0/24
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class C)

Question 2		10.0.0.0/8
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class A)

Question 3		172.16.0.0/16
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class B)

Question 4		192.168.10.0/26
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class C)

Question 5		10.10.0.0/16
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class A)

Question 6		172.20.0.0/20
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class B)

Question 7		200.100.50.0/28
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	

3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class C)

Question 8		15.0.0.0/8
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class A)

Question 9		150.50.0.0/16
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class B)

Question 10		192.168.5.128/25
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	

7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class C)

Question 11		172.31.0.0/12
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class B)

Question 12		100.64.0.0/10
#	Find the following	Answer
1	Network Address	
2	Subnet Mask	
3	Host bits available	
4	Total addresses	
5	Usable hosts	
6	First usable host IP	
7	Last usable host IP	
8	Broadcast address IP	

(Address belongs to Class A)

Answer Key / Solutions

The table below provides complete solutions for all questions.

Q	Given	Class	Network Addr.	Subnet Mask	Host Bits	Total Addresses	Usable Hosts	First Usable Host	Last Usable Host	Broadcast Addr.
1	192.168.1.0/24	C	192.168.1.0	255.255.255.0	8	256	254	192.168.1.1	192.168.1.254	192.168.1.255
2	10.0.0.0/8	A	10.0.0.0	255.0.0.0	24	16,777,216	16,777,214	10.0.0.1	10.255.255.254	10.255.255.255
3	172.16.0.0/16	B	172.16.0.0	255.255.0.0	16	65,536	65,534	172.16.0.1	172.16.255.254	172.16.255.255
4	192.168.10.0/26	C	192.168.10.0	255.255.255.192	6	64	62	192.168.10.1	192.168.10.62	192.168.10.63
5	10.10.0.0/16	A	10.10.0.0	255.255.0.0	16	65,536	65,534	10.10.0.1	10.10.255.254	10.10.255.255
6	172.20.0.0/20	B	172.20.0.0	255.255.240.0	12	4,096	4,094	172.20.0.1	172.20.15.254	172.20.15.255
7	200.100.50.0/28	C	200.100.50.0	255.255.255.240	4	16	14	200.100.50.1	200.100.50.14	200.100.50.15
8	15.0.0.0/8	A	15.0.0.0	255.0.0.0	24	16,777,216	16,777,214	15.0.0.1	15.255.255.254	15.255.255.255
9	150.50.0.0/16	B	150.50.0.0	255.255.0.0	16	65,536	65,534	150.50.0.1	150.50.255.254	150.50.255.255
10	192.168.5.128/25	C	192.168.5.128	255.255.255.128	7	128	126	192.168.5.129	192.168.5.254	192.168.5.255
11	172.31.0.0/12	B	172.16.0.0	255.240.0.0	20	1,048,576	1,048,574	172.16.0.1	172.31.255.254	172.31.255.255
12	100.64.0.0/10	A	100.64.0.0	255.192.0.0	22	4,194,304	4,194,302	100.64.0.1	100.127.255.254	100.127.255.255

Quick reference — Default classful ranges: Class A: 1.0.0.0 – 126.255.255.255 (default mask /8, 255.0.0.0) | Class B: 128.0.0.0 – 191.255.255.255 (default mask /16, 255.255.0.0) | Class C: 192.0.0.0 – 223.255.255.255 (default mask /24, 255.255.255.0). Formulas used: Total addresses = 2^(host bits); Usable hosts = Total addresses – 2 (network & broadcast reserved, for prefixes /30 and shorter).